

Fine-tuning of Medical Reasoning LM

1. Problem Definition

The goal is to fine-tune a base model on a high-quality medical reasoning dataset so the model learns to produce step-by-step reasoning along with the correct answer. The dataset, provided by OpenMed, contains user and assistant chats in a content, reasoning_content, and role format providing step-by-step reasoning. The model must learn to (1): Generate chain-of-thought steps reasoning based on the medical knowledge. (2) Provide the final answer consistent with the reasoning. (3) Goal of maximizing interpretability

2. Track Selection

Track A: Full Reasoning Generation

Input: Medical query

Output: Step-by-step reasoning + final answer

Goal: Maximize interpretability

Track B: Answer-Only Model

Input: Query

Output: Final answer only

Goal: faster inference

Track C: Conversational Medical Assistant

Multi-turn interaction

Maintain context across queries

I chose Track A because this is the track where the model performs reasoning and provides interpretability, which is important in medical domain. Since, the training was supposed to be done on Experiment 1: With Reasoning and Experiment 2: Without Reasoning, Track A is the approach that covers the reasoning part.

3. Model and Finetuning method

SmoLM3-3B

- Even though SmoLM3-3B is a small model, it offers state-of-the-art (SoTA) performance within the small-scale (3B parameter) class, delivering performance that rivals larger 4B models.

- It is trained on a staged curriculum emphasizing reasoning data, and math (GSM8K).
- Compute efficiently with the QLoRA finetuning method and truncated CoT

QLoRA

- With QLoRA (4-bit) and applying low-rank adaptation in the finetuning, trainable model parameters will be significantly less
- All params: 3,082,766,336 || Trainable params: 7,667,712 || trainable parameter percentage: 0.2487%

4. Input-Output Format

Field	Value in dataset
Question	messages[0]['content'] (user query)
Reasoning	messages[1]['reasoning_content'] (assistant reasoning - truncated)
Answer	messages[1]['content'] (assistant content)

Prompt Templates

Experiment 1: With Reasoning

```
<|system|> You are a medical reasoning assistant.
<|user|> {question}
<|assistant|> <reasoning> {short_cot} </reasoning>
Final Answer: {answer}
```

Experiment 1: Without Reasoning

```
<|system|> You are a medical assistant.
<|user|> {question}
<|assistant|> Final Answer: {answer}
```

5. Risks (Medical Safety)

- Hallucination Risk: Fabricated facts, incorrect reasoning, and overconfidence.
- Misinformation: The finetuned model is not a certified medical device.

6. Output Quality

- The validation loss of the model with reasoning is **0.976**, and without reasoning is **0.848**.

Exact Match (EM)	0.5%
BERTScore	0.83
ROUGE-L	0.05
Avg. Reasoning steps	0.4

- The model clearly performs very poorly, since the data is too complex for the architecture, and the maximum token length.